



EMISSIONS REPORT

EMISSIONS REDUCTION RECOMMENDATIONS –
COMPUTING FOCUS

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MERIDAN TRUST BANK

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INTRODUCTION

The purpose of this report is to provide a set of recommendations to reduce the monthly carbon emissions of Meridian Trust Bank (MTB). A specific focus is presented in this document towards reducing ICT emissions.

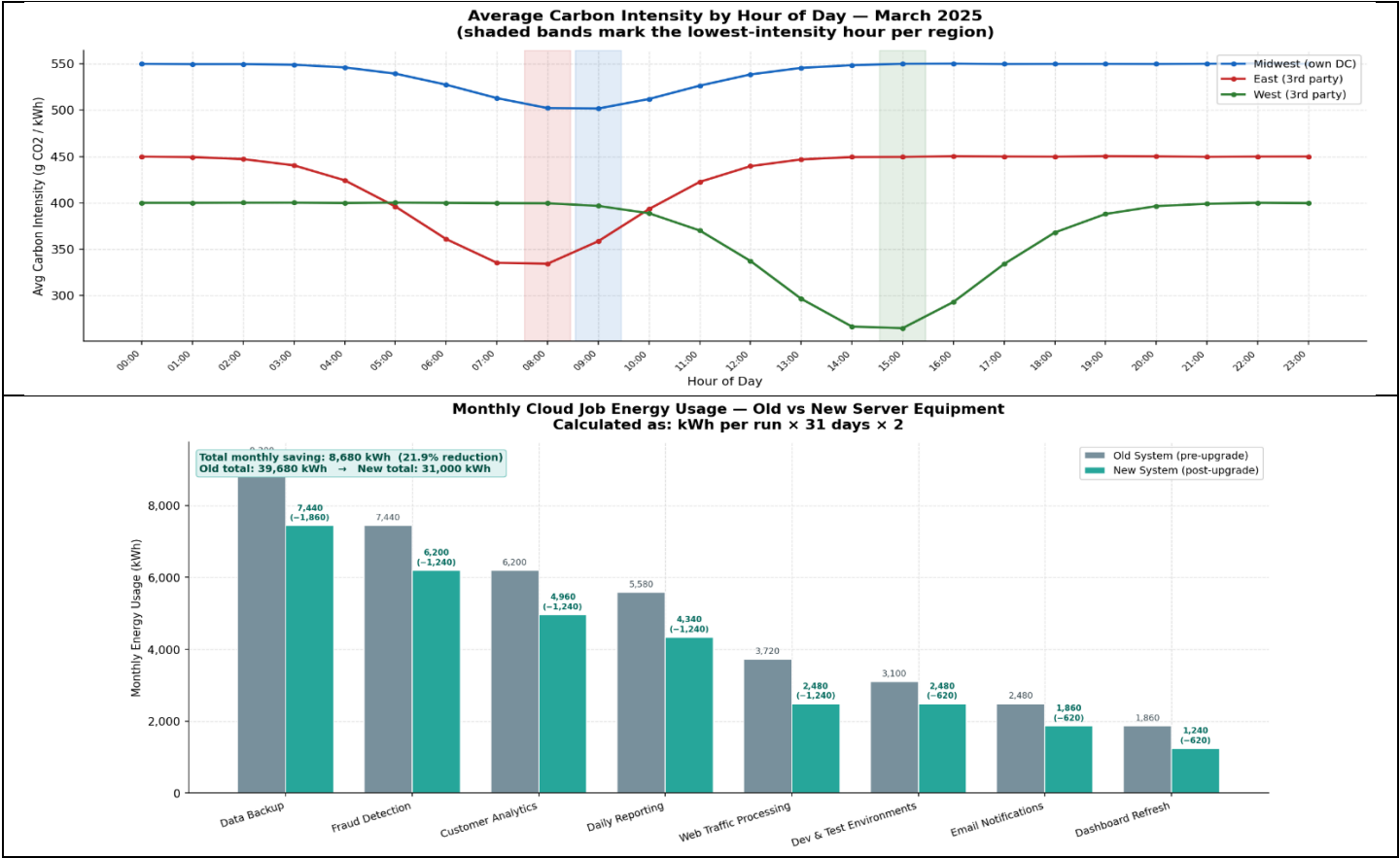
Page 2 of this document outlines zero-cost recommendations for reducing the emissions of MTB's ICT sector. Certain recommendations, such as shifting the time and location of cloud jobs, are zero-cost and can reduce emissions by over 10%. Other recommendations have associated costs but remain well below the allocated budget. All recommendations follow the legal standards outlined in MTB's data privacy policies.

Page 3 of this document focuses on the next steps for MTB to take to improve its standing on the Green Software Foundation's Green Software Maturity Matrix (GSMM). The GSMM is a widely used tool designed to rate organizations on their progress towards fully green practices and to provide guidance on the best next steps. Implementing these changes will position MTB between Levels 2 and 3, well above the rest of the industry.

ZERO-COST REDUCTIONS

The table below provides recommendations for optimal regions and times to perform various daily data processing and cloud tasks. By introducing demand and time shifting, MTB can reduce carbon emissions by almost **800 kg/month** for **zero cost**. Jobs involving restricted data as outlined in MTB’s Data Privacy Policy (MT-SEC-004) are performed on-site as per legal requirements. Modifying the times and locations of daily cloud tasks reduces carbon emissions because, as seen in the first chart, carbon intensity of electricity varies over time and region due to different regions using different energy sources, such as oil or wind, throughout the day.

Job	Opt. Region	Opt. Time	Carbon Reduction (kg)
Fraud Detection	Midwest DC1	8:00-10:00	147.4
Data Backup	Midwest DC2	8:00-10:00	116.8
Customer Analytics	Midwest DC1	10:00-12:00	76.0
Daily Reporting	Midwest DC1	6:00-8:00	65.1
Web Traffic	West	14:00-16:00	166.0
Test Environments	West	14:00-16:00	3.8
Email Notifications	East	8:00-10:00	96.3
Dashboard Refresh	East	8:00-10:00	124.3
Total	-	-	795.9



FURTHER RECOMMENDATIONS

ICT COST-EFFECTIVE REDUCTIONS

As part of this report, the effectiveness of various hardware options to improve the energy efficiency of MTB’s on-site data centers were examined. A budget of \$60,000 was allocated. This report recommends purchasing 11 “Option A” low-cost servers for \$4,000 each, in addition to a new water-cooling system for \$5,000. This requires a total budget of \$49,000, which is \$11,000 below the total allocated budget.

At least 9 low-cost servers are required to complete the daily cloud jobs that MTB must perform on-site per MTB’s privacy policy, and 2 additional servers are recommended for redundancy. Low-cost servers are recommended to help perform server right-sizing as recommended by the GSMM. This will reduce carbon emissions and reduce electricity costs. When servers are not in use, they should be automatically turned off. Purchasing the water-cooling system will also decrease electricity usage and costs by approximately 10%, as detailed in the lower chart on page 2 of this document.

GREEN SOFTWARE MATURITY MATRIX (GSMM)

The table below details MTB’s current state on the industry-standard GSMM. Red squares indicate MTB’s current positioning on the GSMM, green squares indicate MTB’s recommended future position on the chart, and blue squares indicate that MTB is already at the recommended position. Information on the GSMM is detailed in the introduction section of the document. By implementing the changes recommended in this document, MTB can improve to between **Level 2** and **Level 3**, far above the rest of the financial and banking industry.

In addition to the previously stated recommendations in this document, to improve to Level 2 on the GSMM, MTB should provide advanced training in green technology to upper-level management and engineers. In addition, it is recommended for MTB to make a pledge to net zero for Scope 1, 2, and 3 emissions by 2040. Finally, MTB should ensure end-user device carbon waste is minimized by ensuring out-of-date hardware is supported, while continuing to update existing hardware in data centers. These solutions can be performed with minimal cost to MTB, while substantially improving MTB’s public outlook and carbon and cost efficiencies.

	ASPIRING	AWARE	ACTING	AWESOME	INSPIRING
 commitments	none	carbon neutral	carbon zero with offsets	10% (offset)	1% (offset)
 footprint	unknown	know scope 1&2	reducing per unit	reducing absolutely	~zero
 metrics	none	report scope 1&2	daily scope 1&2&3	realtime	predicted
 carbon ops	none	manual	lightswitch ops	auto-rightsizing	carbon SRE
 energy	none	green hosting	dynamic management	demand shaping	24/7 Carbon Free Electricity
 devices	none	some targets	10y/90%	10y/100%	rolling repair
 utilization	none	some multi-tenant	all multi-tenant	max orchestration	edge integration
 products	none	carbon awareness	demand shaping	feature tracking	feature carbon error budgets
 training	ad hoc	basic/champions	advanced	you are the trainer	you are the leader